Chemical Drivers of Summertime Ozone Production in Salt Lake City

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Abstract

In recent years, regular summertime surface ozone (O_3) exceedances of the National Ambient Air Quality Standards have been threatening respiratory and ecosystem health around Salt Lake City and the Northern Wasatch Front (NWF). Despite legislation to control emissions of ozone precursors, measurements across the NWF have not shown effective decreases in surface O_3 and even show increases in some areas. In support of its new State Implementation Plan to understand the problem and identify solutions, the Utah Division of Air Quality performed regional simulations for ozone with the Comprehensive Air Quality Model and Carbon Bond chemical mechanism. They found that compared with observations, their model predictions were underestimating ozone concentrations by approximately 12%. We hypothesize that a combination of poor representations of the underlying chemistry in their model and poor emissions estimates could be responsible for the discrepancies. To test if differences in the underlying chemistry are significantly affecting prediction of O_3 sensitivity to reductions in emissions, we compare modeled ozone by running box model simulations using the Framework for 0-D Atmospheric Modeling (F0AM) Box Model and analyze how an updated chemical mechanism may bring modeled ozone estimates closer to observations. By reducing uncertainties in the chemistry of ozone formation, we hope to improve tools for determining effective strategies that will lower summertime ozone across the NWF.

Background & Motivation

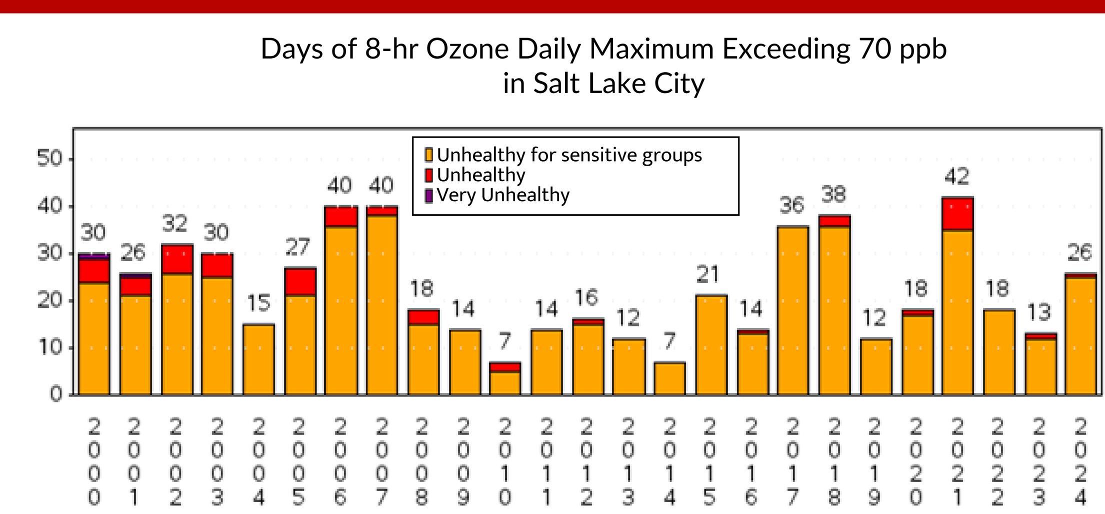


Figure 1: Number of days that ozone exceeded 70 ppb in Salt Lake City, from 2000-2024[1]

- Environmental Protection Agency (EPA) ozone Non-attainment classification for the Northern Wasatch Front (NWF):
  Marginal → Moderate (2015) → Severe (Expected, 2025)[2]

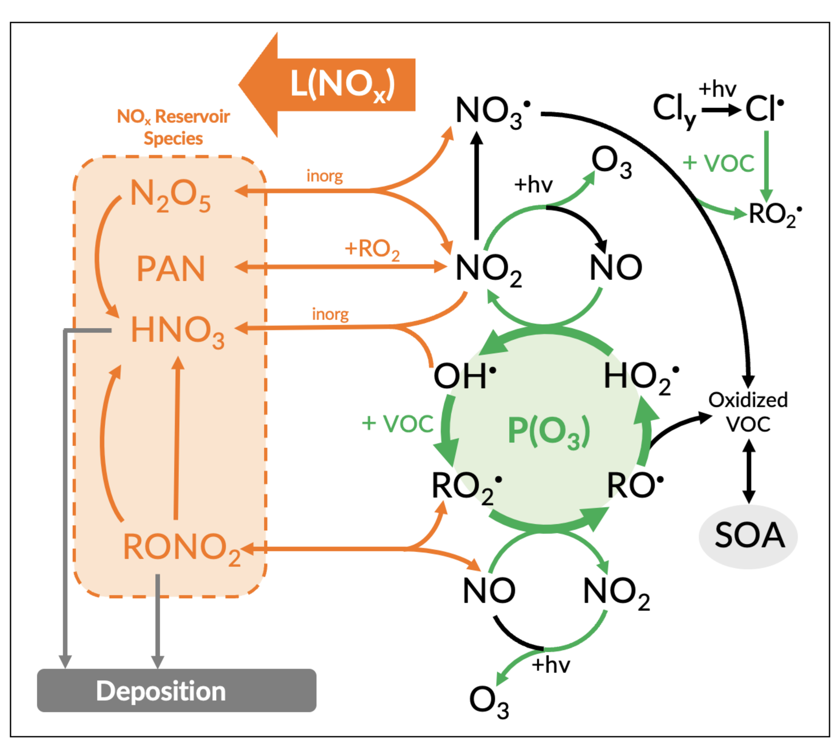


Figure 2: Tropospheric chemistry relevant to the production of ozone. In orange are the loss processes for NO_x. In green are the processes that lead to the production of O_3.

- Non-linear balance between 2 main ozone (O_3) precursors modulate the conditions for efficient ozone production
  - Nitrogen Oxides (NO_x = NO + NO_2)
  - Volatile Organic Compounds (VOCs)
- Understanding of the VOC composition of Northern Wasatch Front is limited, from lack of measurements

Ozone Production Efficiency

$$OPE = \frac{\text{Production rate of } O_3}{\text{Loss rate of } NO_X}$$

In words:
How efficiently a VOC's RO_2 cycles to create ozone (by converting NO into NO_2) before NO_x is permanently lost

- Look at each VOC's OPE to obtain its respective contribution

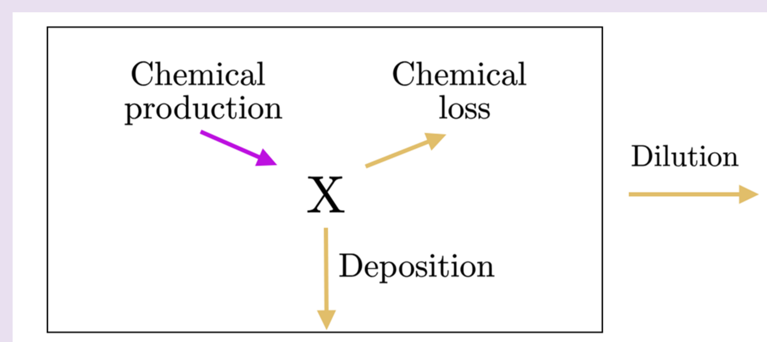
Research Questions

1. **Which VOCs are driving efficient O_3 production in the NWF?**
2. **What differences in the underlying chemistry of various mechanisms affect its prediction of O_3 sensitivity to emissions reductions?**

Methods & Process

USOS Measurements — Constrains gas concentrations at each point
+
Chemical Mechanism — Details chemical reactions & their kinetic rates
→ Run F0AM Model →
Outputs & Results — Concentrations for species in mechanism, production & loss, etc.

MODEL: Framework for 0-D Atmospheric Modeling (F0AM)[3]



- Simulates the atmosphere as well-mixed air within a box (representing a singular point in space) to determine the composition of chemical species at that point over time

Figure 3: How a chemical species, X, changes in F0AM, with sources that increase X (purple) and sinks that decrease X (yellow).[4]

INPUT: 2024 Utah Summertime Ozone Study (USOS) Measurements

- Gas and VOC measurements taken by NOAA Chemical Sciences Laboratory Mobile Van at Hawthorne Elementary School from Jul. 15 – Aug. 18, 2024
- Primary emitted species held constant through each time step, ozone and NO_x evolve in response to given concentrations & relevant reactions

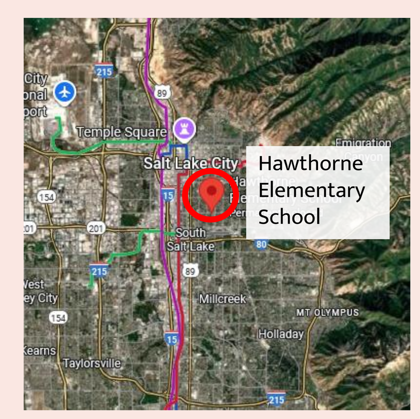


Figure 4: Google Maps view of Hawthorne Elementary School location Salt Lake Valley

Figure 5: Vanessa in front of the van's proton-transfer-reaction (PTR) mass spectrometer, which measured the aromatics

INPUT: Chemical Mechanism

- Describes chemistry: chemical species, how they react, how quickly they react
- Perform box model runs with different mechanisms to see how their output differs

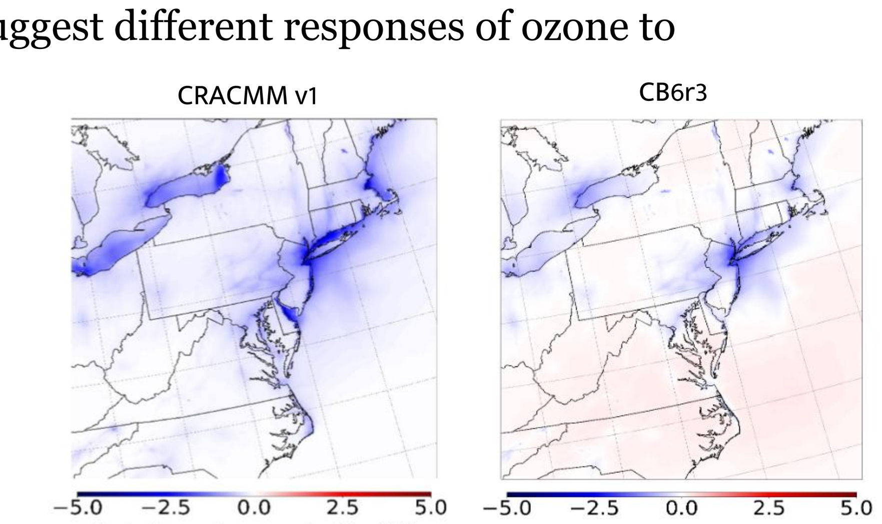
Chemical complexity & realism (− to +)

Mechanism & Version	Updated yield of organic nitrates from aromatics	Includes halogens?	Species & Reactions	Degree of Lumping
RACM 2 (B-VCP)	Unclear	No	132 species 388 rxns	Unclear
Carbon Bond (CB) 6r5h	No	Yes – partial	125 species 329 rxns	Lumped
CRACMM 3	Yes	Yes - full	248 species 921 rxns	Lumped but on reactivity
GEOS-Chem 14.5.2	Yes	Yes – full (state of the science)	326 species 945 rxns	Some lumping
MCM 331	No	No	5832 species 17224 rxns	Explicit

* Results on this poster only for selected mechanisms

Why is the choice of chemical mechanism significant to making policy on emissions reductions?

- Changing only the mechanism can suggest different responses of ozone to emissions policies
- Previous study in Northeast US shows how only CRACMM suggests lowering benzene, toluene, and xylene emissions would reduce O_3 in all areas
- We want to make sure we're choosing the most representative mechanism that suggests the correct species & sources to reduce

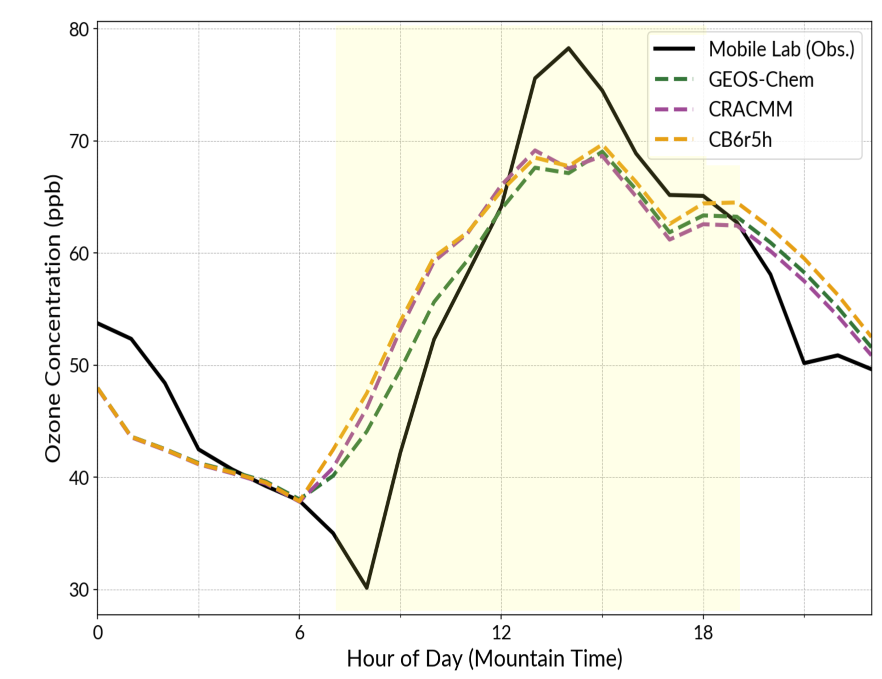


Figures 6a and 6b: From Place et al., 2023;[5] showing the relative change in O_3 in simulations where BTX emissions are zeroed out

Preliminary Results

Very preliminary analysis, constraints still being adjusted in model & data only represent single day

Ozone & OPE comparison between chemical mechanisms



Mechanism	Daytime Mean Bias	Daytime Absolute Error
GEOS-Chem	-0.04 ppb	10.4 ppb
CRACCM	2.39 ppb	12.69 ppb
CB6r5h	3.56 ppb	12.75 ppb

Stats for 9am - 7pm

Daytime Mean Bias: "GEOS-Chem predicts 0.04 ppb lower than the observed ozone."

Daytime Absolute Error: "The average size of the error for GEOS-Chem from the observations is 10.4 ppb."

Figure 7: ozone concentration for each chemical mechanism predicted by F0AM box modeling for July 15, 2024, compared to the observations

- GEOS-Chem & CRACMM have very similar shapes
- CB6r5h deviates from GEOS-Chem & CRACMM most significantly in the afternoon

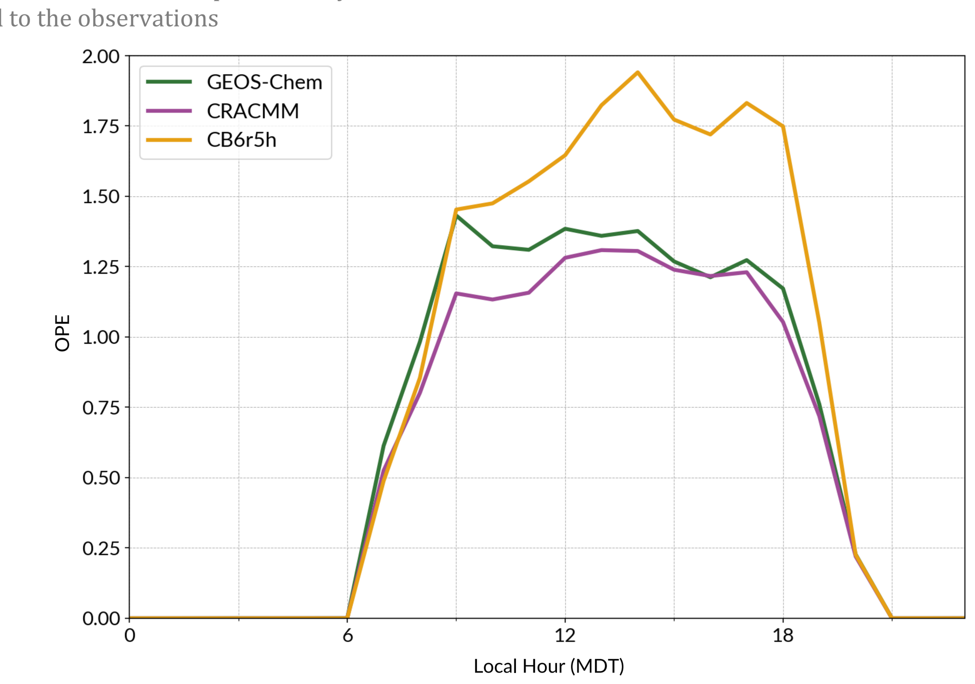
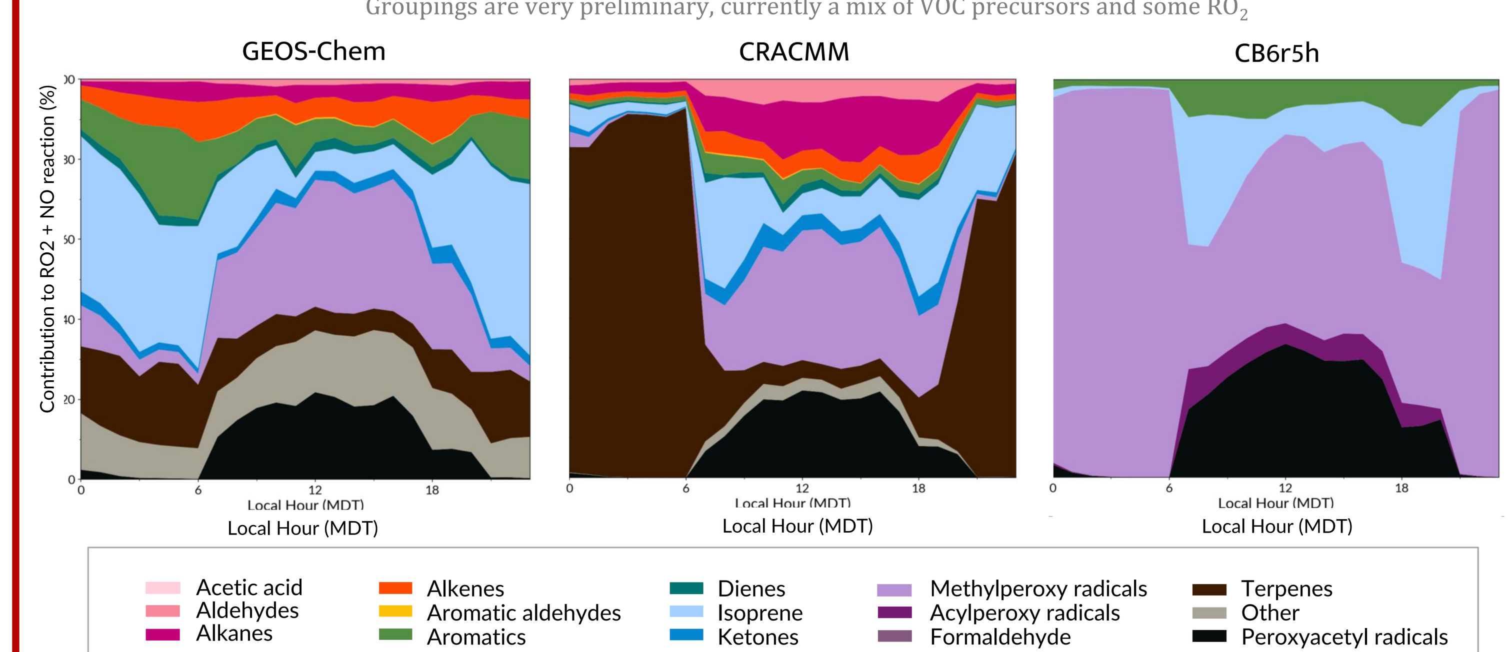


Figure 8: Total OPE for each chemical mechanism predicted by F0AM for July 15, 2024. Accounts for all species that contribute to ozone production and NOx loss.

VOCs most likely to impact ozone production

Groupings are very preliminary, currently a mix of VOC precursors and some RO_2



Figures 9a, 9b, 9c: Contribution of VOC precursor or RO2 to total rate of ozone production from RO2 + NO → NO_2 reaction for each chemical mechanism, predicted by F0AM for July 15, 2024

Future Work

- Explore impact of halogens on O_3 formation: Likely to be small but non-negligible (5-10%)
- Run F0AM simulations for all days of campaign
- Analyze individual OPE for different precursor species
- Add other chemical mechanisms (RACM, MCM) to comparisons

References & Acknowledgements

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